PurpleAgents

A Dual-Reasoning Agentic Framework for Intelligent Purple-Team Cyber Defense

A Project Report

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by

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CERTIFICATE

This is to certify that the report entitled **PurpleAgents: A Dual-Reasoning Agentic Framework for Intelligent Purple-Team Cyber Defense** submitted by **ANJIMA REMANAN** (KNP23AD010), **ROHITH CHANDRAN** (LKNP23AD042) and **ASHID N** (LKNP23AD041), to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the project report **PurpleAgents: A Dual-Reasoning Agentic Framework for Intelligent Purple-Team Cyber Defense**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Dr JYOTHI R L.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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ABSTRACT

PurpleAgents is an autonomous *purple-team* framework for testing and strengthening the safety of AI systems such as large language models (LLMs), chatbots, and AI agents. Traditional security work is split: attackers (red teams) test a system only now and then, defenders (blue teams) react after something goes wrong, and the lessons from each test are rarely turned into lasting protection. PurpleAgents brings attack and defense together into one continuous, self-improving cycle inside a safe, authorized environment.

The core of the framework is a **dual-reasoning engine**. For every attack, the engine reaches a conclusion in two independent ways. The first path is **evidence-first**: it looks at what actually happened — the prompt sent and the model's response — and decides whether the attack succeeded. The second path is **hypothesis-first**: it takes a list of known attack types and tests the evidence against each one, keeping only the explanation that fits. A conclusion is accepted only when both paths agree. This prevents the system from trusting a single wrong signal.

An offensive (red) agent sends adversarial prompts such as jailbreaks and prompt injections to the target model, while a defensive (blue) agent tries to detect and block unsafe behaviour. A reflection step then studies each missed attack, finds the exact reason it slipped through, and proposes a concrete fix such as a stronger guardrail rule. Every attack, its evidence, the reasoning path, the confidence level, and the outcome are stored in an auditable memory. The defining idea is a **closed learning loop**: a missed attack is not just a failure — it becomes a verified lesson, so the defense improves on the next cycle. Human approval governs any high-impact change. By attacking, defending, reasoning, and learning in one loop, PurpleAgents aims to make an AI system measurably harder to break over time.

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LIST OF SYMBOLS

Symbol	Meaning
ASR	Attack Success Rate (fraction of attacks that got through)
DR	Detection rate / catch-rate, $\mathrm{DR} = 1 - \mathrm{ASR}$
E	Evidence bundle {prompt, response, refused/complied, guardrail signal}
H	Hypothesis set (list of known attack categories)
A	Evidence-first conclusion
B	Hypothesis-first conclusion
C	Confidence level of an accepted conclusion
r	Round index
ASR_r	Attack success rate at round r
θ	Agreement / convergence threshold
$\Delta\mathrm{DR}$	Improvement in detection rate relative to round 1, $\mathrm{DR}_r - \mathrm{DR}_1$
FFR	False-flag rate (fraction of harmless prompts wrongly blocked)

Abbreviation	Expansion
LLM	Large Language Model
CLI	Command-Line Interface
SOC	Security Operations Centre
OWASP	Open Worldwide Application Security Project
ATLAS	Adversarial Threat Landscape for AI Systems (MITRE)

CHAPTER 1

INTRODUCTION

AI systems such as chatbots and agents built on large language models (LLMs) are now used almost everywhere. They answer customer questions, help people write code, power search, and automate everyday tasks. As these systems spread, so do the ways they can be misused. An attacker can trick a model into producing unsafe or confidential output by using carefully written prompts. Two common examples are *jailbreak* prompts, which make a model ignore its safety rules, and *prompt injection*, where hidden instructions take control of the model's behaviour.

Keeping these systems safe is hard because the threats keep changing. A one-time safety check is not enough. New attack tricks appear all the time, and a model that was safe last month may become unsafe after an update.

The usual approach to this problem is fragmented. Red teams (attackers) test for weaknesses only now and then. Blue teams (defenders) respond after an alert has been raised. The lessons learned from each test are not reliably turned into stronger defense. The two sides rarely work together inside one system that keeps improving on its own.

PurpleAgents is proposed to fix this. It is an autonomous *purple-team* framework that continuously attacks and defends a target AI model inside a safe, authorized environment. It checks every conclusion using two independent lines of reasoning, and it learns from every missed attack so that the defense keeps getting better.

1.1 Motivation

Three trends make this project timely.

- **AI is now a front door.** Many organisations place a chatbot or an AI agent directly in front of their customers, staff, or internal data. A single successful jailbreak can expose private information, produce harmful content under the organisation's name, or make an agent take an action it should not. The cost of a missed attack is therefore real and immediate.
- **Attacks are getting cheaper.** Recent research shows that LLM-based agents can plan and carry out attacks on their own [9, 24]. When attacks can be automated, defense that depends on occasional human testing falls behind. Defense must also become continuous and automatic.
- **Single signals are unreliable.** A safety filter or an LLM "judge" can be wrong.

If a system trusts one signal, it can miss an attack or block a harmless request without anyone noticing. Checking every decision in two independent ways is a simple, practical way to catch such mistakes.

The literature already describes the idea of a purple agent that passes lessons between attacker and defender [25], but it does not build one with verified reasoning and a learning loop. This gap, together with the trends above, is the motivation for PurpleAgents.

1.2 Existing System

Today's AI-safety and testing tools each cover only one part of the problem.

- **Red-team / attack tools** can automatically generate jailbreaks and injection prompts. However, they only *find* weaknesses. They do not defend the system, and they do not learn from what they find.
- **Defensive tools** such as guardrails and output filters can block some unsafe responses. However, they react to single signals and are often bypassed when an attacker uses new phrasing.
- **Manual purple teaming** brings humans into the loop. However, it is slow, happens only periodically, and depends heavily on expert effort.

In summary, existing systems suffer from the following problems:

- **Fragmentation:** attack and defense are kept separate, so lessons are not shared between them.
- **Single-path decisions:** one alert or one score is trusted, so a wrong call or a missed attack goes unchecked.
- **No structured learning:** a missed attack is logged as a failure, not turned into a reusable lesson.
- **Periodic, manual effort:** testing happens occasionally, not continuously.

1.3 Problem Statement

There is a need for an intelligent system that can continuously attack an AI model, defend it, verify its own conclusions, and learn from every missed attack — all on its own, in a safe and authorized environment. Current tools cannot do this. They solve only one piece of the problem (attack, or defense, or reasoning), they rely on a single reasoning path, and they do not convert failures into verified, reusable lessons. This gap leaves AI systems exposed to new and evolving attacks.

1.4 Objectives

The objectives of this project are:

1. To build an autonomous purple-team framework in which attack and defense agents run continuously against a target AI model.
2. To introduce a **dual-reasoning engine**, so that a conclusion is accepted only when an evidence-first path and a hypothesis-first path agree.
3. To detect attacks and, for every missed attack, find the exact reason it was missed.
4. To generate testable defensive improvements, such as stronger guardrail rules.
5. To keep an auditable memory of evidence, reasoning, confidence, and outcome.
6. To close the loop, so that each missed attack becomes a verified lesson that strengthens the defense, with human approval on key changes.
7. To show, through repeated attack–defend cycles, that the defense measurably improves over time.

1.5 Applications

PurpleAgents is intended for any organisation that builds, deploys, or protects AI systems. The main users are:

- **Companies deploying chatbots and AI assistants.** Banks, online stores, hospitals, and service providers that put a chatbot in front of their customers can use PurpleAgents to test it continuously before and after each update, and to strengthen its guardrails over time.
- **AI and LLM providers.** Teams that build or fine-tune language models can use the framework as an automated safety check during development. A model can be attacked in repeated rounds, and the lessons can be turned into better safety rules before release.
- **Security Operations Centre (SOC) teams.** SOC analysts who are responsible for AI systems can use the auditable memory as a clear record of what was attacked, how it was judged, and what was fixed. Low-confidence cases are sent to them for review instead of being decided silently.
- **Developers of AI agents.** Agents that read web pages, e-mails, or documents are exposed to prompt injection. PurpleAgents can test such agents against injection attacks inside a safe environment.

- **Education and research.** The framework can be used in teaching and research to show how attack, defense, and reasoning work together, and to compare different guardrail strategies in a fair, repeatable way.

1.6 Organization of the Report

The rest of this report is organized as follows. Chapter 2 reviews related research on autonomous attackers, autonomous defenders, and reasoning methods. Chapter 3 defines the problem and the scope of the project. Chapter 4 lists the functional, non-functional, hardware, and software requirements. Chapter 5 presents the system design, the proposed architecture, the dual-reasoning engine, and the algorithm. Chapter 6 covers the tools, the data sources, and the evaluation plan. Chapter 7 presents the feasibility study and the project timeline, and Chapter 8 states the expected outcomes and concludes the report.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Recent research covers three areas that are relevant to this work: autonomous attackers (red), autonomous defenders (blue), and reasoning and learning methods. The works reviewed below show that each area is advancing quickly. However, no single system joins them into one self-improving loop. All 25 papers were read and summarised from their abstracts. They are grouped into six themes, starting with the works closest to PurpleAgents.

2.2 Autonomous and Agentic Purple Teaming

2.2.1 Toward an Autonomous Purple Teaming Framework for Security and Safety in LLMs (AutoPT)

L. H. Park et al., PRDC 2025 [1]

This work runs two co-evolving agents in a loop. A red module uses coverage-guided fuzzing to find jailbreaks, and a blue module self-heals its guardrails using that feedback. It is the closest published match to our idea. **Gap:** it covers LLM safety only and relies on a single measurement score, with no dual reasoning.

2.2.2 Purple-teaming LLMs with Adversarial Defender Training (PAD)

A. Zhou et al., arXiv:2407.01850, 2024 [2]

PAD trains an attacker and a defender together, so that the defender learns to catch and explain unsafe responses. We cite it because it shows that red/blue co-training works. **Gap:** it focuses on guardrail training only and does not form a closed loop.

2.2.3 From Promise to Peril: Rethinking Red and Blue Teaming in the Age of LLMs

A. Abuadbba et al., arXiv:2506.13434, 2025 [3]

This paper maps the use of LLMs in red and blue teaming onto known security frameworks and points out weaknesses such as hallucination. It supports our choice to keep humans in the loop. **Gap:** it is a position paper; no system is built.

2.3 Offensive (Red) Agents

2.3.1 VulnBot: Autonomous Penetration Testing for a Multi-Agent Collaborative Framework

H. Kong et al., arXiv:2501.13411, 2025 [4]

VulnBot is a multi-agent LLM system that splits a penetration test into reconnaissance, scanning, and exploitation. It gives us a template for structuring our red agents. **Gap:** it covers attack only, with no defense or learning.

2.3.2 RedTeamLLM: An Agentic AI Framework for Offensive Security

B. Challita and P. Parrend, arXiv:2505.06913 / IFIP AI4S 2025 [5]

RedTeamLLM is an agentic penetration-testing system built on a summarize–reason–act loop. It addresses plan correction and memory. It informs the reasoning and memory design of our red side. **Gap:** it is offensive only, with no validation step and no memory of lessons learned.

2.3.3 Autonomous Adversary: Red-Teaming in the Age of LLM

A. K. M. Mamun et al., arXiv:2605.06486, 2026 [6]

This work benchmarks LLM-driven attacks and uses an LLM as a judge to check whether each attack step succeeded. It informs how our validation step decides success. **Gap:** it is a red-side benchmark with no defense loop.

2.3.4 PentestGPT: Evaluating and Harnessing LLMs for Automated Penetration Testing

G. Deng et al., USENIX Security 2024, arXiv:2308.06782 [7]

PentestGPT splits the work into three cooperating modules so that the model does not lose track of the test. It is a foundation for our modular red agent. **Gap:** it is a penetration-testing tool with single-path reasoning and no blue side.

2.3.5 AutoAttacker: An LLM-Guided System to Implement Automatic Cyber-attacks

J. Xu et al., arXiv:2403.01038, 2024 [8]

AutoAttacker focuses on the post-breach, "hands-on-keyboard" stage of an attack. It shows the kind of post-exploitation actions a red agent may need to simulate. **Gap:** it covers post-breach attacks only, with no defense.

2.3.6 LLM Agents Can Autonomously Hack Websites

R. Fang et al., arXiv:2402.06664, 2024 [9]

This work shows that agents built on a leading LLM can perform SQL injection without prior knowledge of the flaw, with a reported success rate of about 73%. It is concrete proof that autonomous attackers are real. **Gap:** it is an attack demonstration with no defense side.

2.3.7 On the Feasibility of Using LLMs to Execute Multistage Network Attacks (Incalmo)

B. Singer et al., arXiv:2501.16466, 2025 [10]

This paper finds that plain LLMs fail at multi-host attacks until an abstraction layer is added (an attack-graph service, an environment-state service, and an action planner). These services resemble our attack-graph reasoning and evidence layer. **Gap:** it addresses attack execution only, with no reasoning or learning loop.

2.4 Defensive and SOC Agents

2.4.1 CORTEX: Collaborative LLM Agents for High-Stakes Alert Triage

B. Wei et al., arXiv:2510.00311, 2025 [11]

In CORTEX, specialised agents collaborate over real evidence: a behaviour-analysis agent, evidence-gathering agents, and a reasoning agent that makes an auditable decision. It is the closest match to our blue side and our evidence-based reasoning. **Gap:** it performs blue-side triage only, with no red side and no loop.

2.4.2 Large Language Models for Security Operations Centers: A Comprehensive Survey

arXiv:2509.10858, 2025 [12]

This survey explains how LLMs fit into the security operations centre (SOC) workflow, including triage, correlation, and prioritisation. It grounds what our defensive agents are expected to do. **Gap:** it is a survey; no system is built.

2.4.3 A Survey of Agentic AI and Cybersecurity: Challenges, Opportunities and Use-case Prototypes

S. J. Lazer et al., arXiv:2601.05293, 2026 [13]

This survey studies agentic AI in security and warns of agent-specific risks such as memory poisoning. It also includes prototypes. It supports our multi-agent design and our human-approval layer. **Gap:** it offers a survey and prototypes, but no closed loop.

2.5 Reinforcement-Learning Cyber Operations

2.5.1 CybORG: An Autonomous Cyber Operations Research Gym

C. Baillie et al., arXiv:2002.10667, 2020 [14]

CybORG is an open simulation "gym" in which red (attacker) and blue (defender) agents are trained together using reinforcement learning. It shows the kind of safe range needed for co-training agents. **Gap:** it is a training environment only, with no reasoning engine.

2.5.2 Deep Reinforcement Learning for Autonomous Cyber Operations: A Survey

C. Palmer et al., arXiv:2310.07745, 2024 [15]

This survey frames cyber defense as a game between a blue defender and a red attacker and lists the main open challenges. It supports our training setup. **Gap:** it is an RL survey with no dual reasoning.

2.5.3 Entity-based Reinforcement Learning for Autonomous Cyber Defence

I. Symes Thompson et al., arXiv:2410.17647, 2024 [16]

This work builds a defensive agent that keeps working when the network changes, by viewing the network as a set of entities. It guides how our defensive agents can stay robust across different systems. **Gap:** it covers defense only and uses single-path decisions.

2.5.4 Hierarchical Multi-agent Reinforcement Learning for Cyber Network Defense

A. Singh et al., arXiv:2410.17351, AAMAS 2025 [17]

This paper splits defense into sub-tasks handled by several agents under a master policy. It is a model for how our blue agents can cooperate. **Gap:** it addresses blue-side coordination only, with no reasoning layer.

2.5.5 Towards Autonomous Cyber Operation Agents: Exploring the Red Case

L. Li et al., arXiv:2309.02247, 2023 [18]

This work trains an attacking red agent with reinforcement learning and studies how to make it work across different networks. It supports the building of our offensive agents. **Gap:** it covers red-agent training only.

2.5.6 A Red Teaming Framework for Evaluating Robustness of AI-enabled SOAR Systems

A. J. Shaikh et al., arXiv:2605.17075, 2026 [19]

This framework combines an LLM planner with a reinforcement-learning controller to run multi-stage attacks against automated defenders. It finds that an LLM alone cannot sustain multi-stage attacks. It is a real example of an attacker-versus-defender loop. **Gap:** it focuses on attack testing only, with no learning loop.

2.6 Dual and Abductive Reasoning

2.6.1 Evidential Cyber Threat Hunting

F. Araujo et al., SIAM AI4CS 2021, arXiv:2104.10319 [20]

This paper proposes a formal reasoning framework that works over three spaces — knowledge, hypothesis, and action — and attaches a confidence level to each conclusion. It is the academic basis of our dual-reasoning engine. **Gap:** it is a reasoning framework only and is not integrated into an attack–defense system.

2.6.2 Diagnosing Unknown Attacks in Smart Homes Using Abductive Reasoning

K. Ramkumar et al., IEEE, arXiv:2412.10738, 2025 [21]

This work combines anomaly detection with abductive reasoning to find the class of an unknown attack and the requirement it violated. It gives a concrete method for our hypothesis-first path. **Gap:** it performs diagnosis only, with no attack–defense loop.

2.7 Surveys and Foundations

2.7.1 When LLMs Meet Cybersecurity: A Systematic Literature Review

J. Zhang et al., Cybersecurity (Springer), 2025, arXiv:2405.03644 [22]

This review covers more than 300 works across 25 LLMs and over 10 security tasks. It is the main peer-reviewed foundation for our related work. **Gap:** it is a broad survey that provides motivation only.

2.7.2 Large Language Models for Cyber Security: A Systematic Literature Review (LLM4Security)

H. Xu et al., arXiv:2405.04760, 2024 [23]

This is a second broad review of LLMs across security tasks. It adds breadth to our related work. **Gap:** it is a task survey and does not propose a framework.

2.7.3 Forewarned is Forearmed: A Survey on LLM-based Agents in Autonomous Cyberattacks

M. Xu et al., arXiv:2505.12786, 2025 [24]

This survey breaks attack agents into scouting, memory, reasoning, and action. It warns of "Cyber Threat Inflation", where attacks become cheaper and larger in scale. It shows why autonomous defense is urgent. **Gap:** it surveys the attacker side; the defense side is missing.

2.7.4 From Texts to Shields: Convergence of LLMs and Cybersecurity

T. Li et al., arXiv:2505.00841, 2025 [25]

This paper directly describes the purple-teaming model: a red agent attacks, a blue agent learns to defend, and a purple agent transfers the lessons between them. It shows that our core idea is already recognised in the literature. **Gap:** it is a survey that describes the idea but does not build it.

2.8 Comparison of Existing Systems

Table 2.1 compares the main groups of existing work with PurpleAgents on five features: whether the system attacks, whether it defends, what kind of reasoning it uses to reach a decision, whether it has a learning loop, and whether it includes human approval. Each row summarises a group of papers, based on our reading of their abstracts. "Not addressed" means the feature is not part of the work's stated contribution.

Table 2.1: Comparison of existing systems with PurpleAgents

System group	Attack	Defense	Reasoning type	Learning loop	Human approval
Autonomous purple teaming [1, 2]	Yes	Yes	Single score or trained classifier	Co-evolution / co-training; no stored lessons	Not addressed
Offensive (red) agents [4, 5, 6, 7, 8, 9, 10]	Yes	No	Single-path LLM planning; LLM-as-judge in [6]	No	Not addressed
Defensive / SOC agents [11, 12, 13]	No	Yes	Multi-agent, evidence-based single decision	No	Recommended in [13]
RL cyber operations [14, 15, 16, 17, 18, 19]	Yes (simulated)	Yes (simulated)	Learned policy (reward-driven)	Reward-based training, not lesson-based	Not addressed
Dual / abductive reasoning [20, 21]	No	Diagnosis only	Hypothesis-based, with confidence	No	Not addressed
Surveys and position papers [3, 22, 23, 24, 25]	Described	Described	Discussed only	Described in [25], not built	Recommended in [3]
PurpleAgents (proposed)	**Yes**	**Yes**	**Dual: evidence-first + hypothesis-first, with convergence check**	**Closed loop of verified, stored lessons**	**Required for high-impact changes**

The table shows the pattern clearly. Each group is strong in one or two columns, but no group covers all five. In particular, no existing group combines dual-path reasoning with a closed loop of stored, verified lessons under human approval.

2.9 Summary

Across these works, attack agents and defense agents are each well developed, and reasoning and learning methods exist separately. However, no existing system joins them into one closed loop that learns from every missed attack. That integrated loop — continuous attack and defense, dual-path reasoning, validation, auditable memory, and learning — is the gap that PurpleAgents fills.

CHAPTER 3

PROBLEM DEFINITION

3.1 Problem Description

AI systems are being protected with methods that were designed for a slower, more predictable world. The main problems are described below.

- **Fragmented security:** attackers (red teams) and defenders (blue teams) work separately. A weakness found by the red team is reported, but it is not automatically turned into a better defense. Likewise, what the blue team learns from an alert rarely shapes the next round of testing. Lessons from one side seldom help the other.
- **Periodic, manual testing:** AI systems are tested only now and then, often by hand. Between two tests, weaknesses stay hidden while new attack tricks keep appearing. A model update can also bring back problems that had already been fixed.
- **Single-path trust:** existing tools depend on one alert, one score, or one model's decision. If that single signal is wrong, nothing checks it. A harmless response may be flagged, or, worse, a successful attack may be marked as safe and go unnoticed.
- **No learning from misses:** when an attack slips through, the system records it as a failure. It does not find out *why* the attack succeeded, and it does not turn that reason into a lasting lesson or a concrete fix.
- **Result:** AI systems remain exposed to new and evolving attacks, and their defenses do not improve on their own.

3.2 Existing System Limitations

Table 3.1 links each limitation of current practice to its effect on the safety of an AI system.

Table 3.1: Limitations of existing systems and their impact

Limitation	Impact
Attack and defense are separate	Weaknesses found by the red team are not turned into defenses automatically; the same attack can succeed again.
Testing is periodic and manual	New attacks and model updates create weaknesses that stay hidden until the next test.
Decisions trust a single signal	A wrong judgement by one filter or one judge goes unchecked, causing missed attacks or wrongly blocked requests.
Misses are logged, not analysed	The root cause of a miss is never found, so the defense does not improve.
Fixes are not re-tested	A new rule may not block the attack, or may break normal use, and nobody notices.
No complete audit trail	It is hard to explain later why a decision was made or why a rule was added.

3.3 Scope

3.3.1 In Scope

- A framework that attacks and defends a target AI model (an LLM, chatbot, or AI agent) through its text API, inside a safe and authorized environment.
- An autonomous red agent (attack) and blue agent (defense).
- A dual-reasoning engine made of an evidence-first path, a hypothesis-first path, and a convergence check.
- A reflection step that finds the cause of a miss and proposes a fix.
- A validation step that re-tests each fix.
- An auditable memory and a closed learning loop.
- Human approval for high-impact changes.
- Evaluation through repeated attack–defend cycles, measuring how the defense improves.

3.3.2 Out of Scope

- Attacking live third-party hosted services. Only a target that the team controls is attacked.
- Training a new foundation model from scratch. A ready-made open-source model is used instead.
- Network, host, or hardware attacks outside AI systems.
- Large-scale commercial deployment and large user studies.

CHAPTER 4

REQUIREMENTS SPECIFICATION

4.1 Functional Requirements

FR1. The system shall connect to a target AI model through its text API, send prompts, and read responses.

FR2. The red agent shall select attack prompts from a prepared set and/or generate new ones, covering known attack categories.

FR3. The blue agent shall detect and block unsafe responses using guardrails and output filters, and shall label each response as safe or unsafe.

FR4. The system shall collect evidence for every attack: the prompt, the model's response, whether the model refused or complied, and any guardrail signal.

FR5. The dual-reasoning engine shall produce an evidence-first conclusion and a hypothesis-first conclusion, and shall accept a result only when both agree. Otherwise, it shall mark the result as low-confidence and flag it for a human.

FR6. For a missed attack, the reflection step shall find the root cause and propose a concrete, testable fix.

FR7. The validation step shall re-test each fix to confirm that it blocks the attack.

FR8. The system shall store evidence, reasoning path, confidence, outcome, and lessons in an auditable memory.

FR9. The learning loop shall apply verified lessons to the next cycle.

FR10. Human approval shall be required before deploying fixes or other high-impact changes.

4.2 Non-Functional Requirements

- **Performance:** the system should complete one attack–reason–validate cycle in a reasonable time.
- **Accuracy:** dual-path reasoning should reduce wrong conclusions compared with a single reasoning path.
- **Reliability:** the system should run repeated cycles without losing memory or state.

- **Usability:** the system should produce clear logs and summaries of what was attacked, what was caught, and what was fixed.
- **Scalability and Extensibility:** the system should work with any target that exposes a text API, and it should be easy to add new attack categories.
- **Security and Privacy:** the system should operate only on an authorized target, use no real user data, and keep all testing inside a controlled environment.

4.3 Hardware Requirements

- Multi-core CPU and 16 GB or more of RAM.
- Optional GPU. This helps when running a local open-source model, but it is not required when using a cloud-hosted model through an API.
- Sufficient storage (an SSD is recommended).
- A reliable internet connection (required when using a cloud-hosted model).

4.4 Software Requirements

- Operating system: Windows or Linux.
- Python 3.10 or higher.
- Standard AI/agent and HTTP libraries.
- Open-source red-teaming and attack-generation tools.
- An open-source guardrail / safety-classifier toolkit (for the blue side).
- Access to an open-source LLM, run locally or through a cloud-hosted inference API.
- PostgreSQL (for the auditable memory and logs).

4.5 Constraints and Assumptions

4.5.1 Constraints

1. Phase 1 targets AI systems (LLM, chatbot, or agent) that are reached through a text API.
2. Attacks are performed only on a target that the team controls and is authorized to test — never on live third-party services.

3. The project uses pre-trained open-source models. It does not train a foundation model from scratch.
4. Evaluation uses a limited set of attack categories and prompts because of time and compute limits.

4.5.2 Assumptions

1. The target model exposes a text input/output API.
2. The attack categories drawn from public sources reasonably cover current threats.
3. A human reviewer is available to approve high-impact changes.

CHAPTER 5

SYSTEM DESIGN

This chapter describes what will be built and how it will work. It starts with an overview of the deliverable, then presents the flow chart, the architecture, the dual-reasoning engine, each module in detail, the algorithm, the interface design, and finally a worked example that follows one attack through the whole loop.

5.1 System Overview

5.1.1 What We Will Build

PurpleAgents will be built as a **Python-based framework** with the following parts:

- A **Red Agent** module that sends attack prompts to a target AI model through its API.
- A **Blue Agent** module that detects and blocks unsafe responses using guardrails.
- A **Dual-Reasoning Engine** that verifies every conclusion using an evidence-first and a hypothesis-first path.
- A **Reflection** module that finds why an attack was missed and proposes a fix.
- A **Validation** module that re-tests fixes.
- An **Auditable Memory** that stores evidence, reasoning, and outcomes.
- A **Closed Learning Loop** that applies verified lessons to the next cycle, under human approval.

The framework is operated through a **command-line interface (CLI)** only. All runs, reviews, and reports are started from the CLI, and results are written to the auditable memory and to plain-text round summaries.

What we do not build. The project does **not** build a new AI model. It uses a ready-made open-source model as the target, and builds the framework around it. The reasoning "judge" is also an existing model that is prompted, not trained. The contribution of the project is the framework: the agents, the dual-reasoning check, the learning loop, and the memory that connects them.

5.1.2 Planned Code Structure

The framework will be organised as one Python package, with one sub-module per part of the system. The planned layout is shown below; file names may change during implementation.

```
purpleagents/
  config.yaml # target endpoint, attack categories, rounds
  cli.py # command-line entry point
  target/client.py # sends prompts to the target model's text API
  red/agent.py # selects, generates and mutates attack prompts
  blue/guardrails.py # input and output guardrail rules
  blue/agent.py # labels each response safe or unsafe
  reasoning/engine.py # evidence-first, hypothesis-first, convergence
  reasoning/hypotheses.yaml # known attack categories (H)
  reflection/reflect.py # root cause and proposed fix
  validation/validate.py # caught/missed check and fix re-testing
  memory/store.py # append-only audit records and lessons
  approval/queue.py # pending fixes and flagged cases for a human
```

5.2 Flow Chart

Figure 5.1 shows the working process of PurpleAgents. The process is a loop, and each pass through the loop is called a *round* or *cycle*.

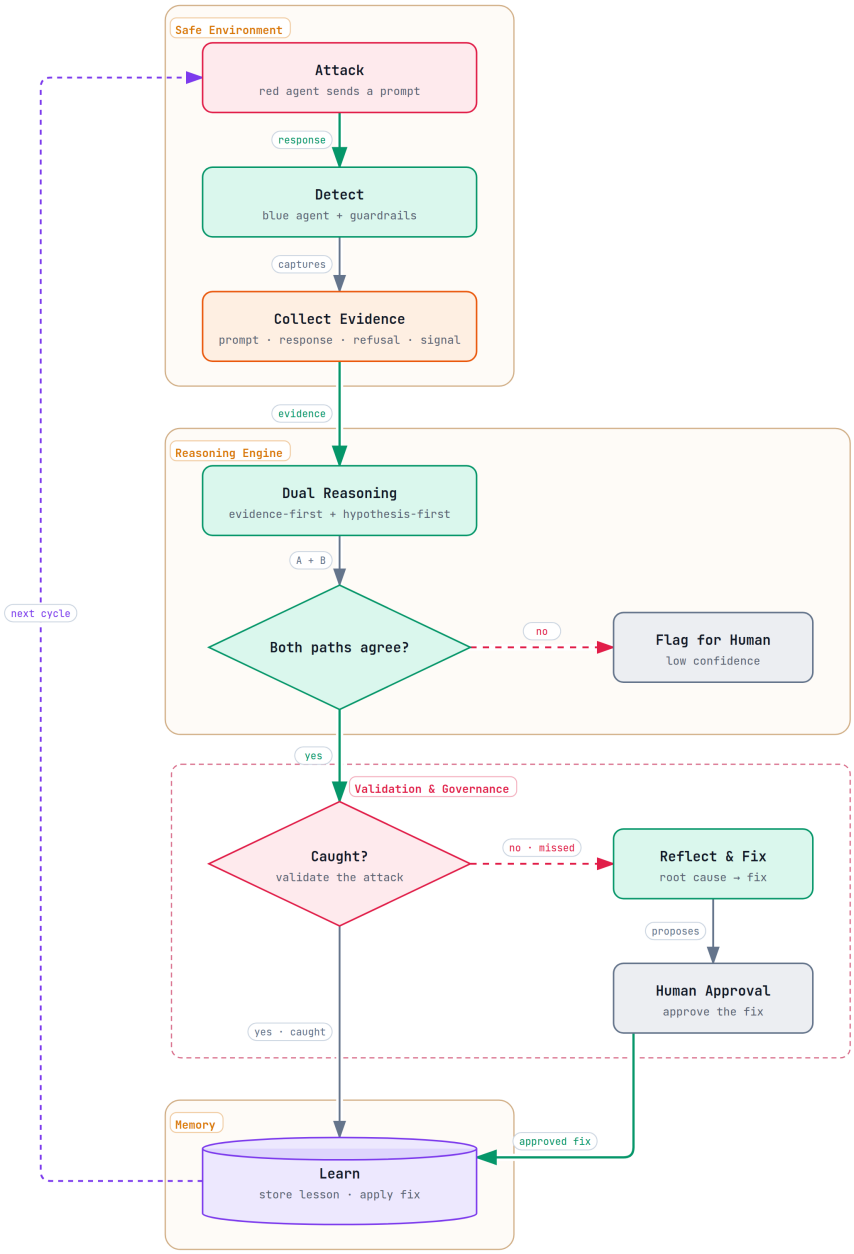


Figure 5.1: Working-process flowchart of PurpleAgents, showing the closed learning loop

The process works as follows. First, the red agent attacks the target model by sending it an adversarial prompt. The blue agent then watches the model's response and any guardrail signals and tries to defend. Next, the evidence is collected: the prompt, the response, whether the model refused or complied, and the guardrail signal.

The dual-reasoning engine studies this evidence and decides what happened. If its two reasoning paths agree, the conclusion is accepted with a confidence level. If they disagree, the case is flagged for a human. The validation step then checks whether the attack was caught. If it was caught, the system moves straight to the learning step. If it was missed, the reflection step finds the root cause and proposes a fix, which a human

must approve.

Finally, the lesson is stored in memory and the approved fix is applied to the defense. The loop then returns to the attack step for the next round. Because every round starts with a stronger defense than the one before, the system is expected to improve over time.

5.3 Proposed Architecture

Figure 5.2 shows the proposed system architecture. Its main blocks and their connections are described below.

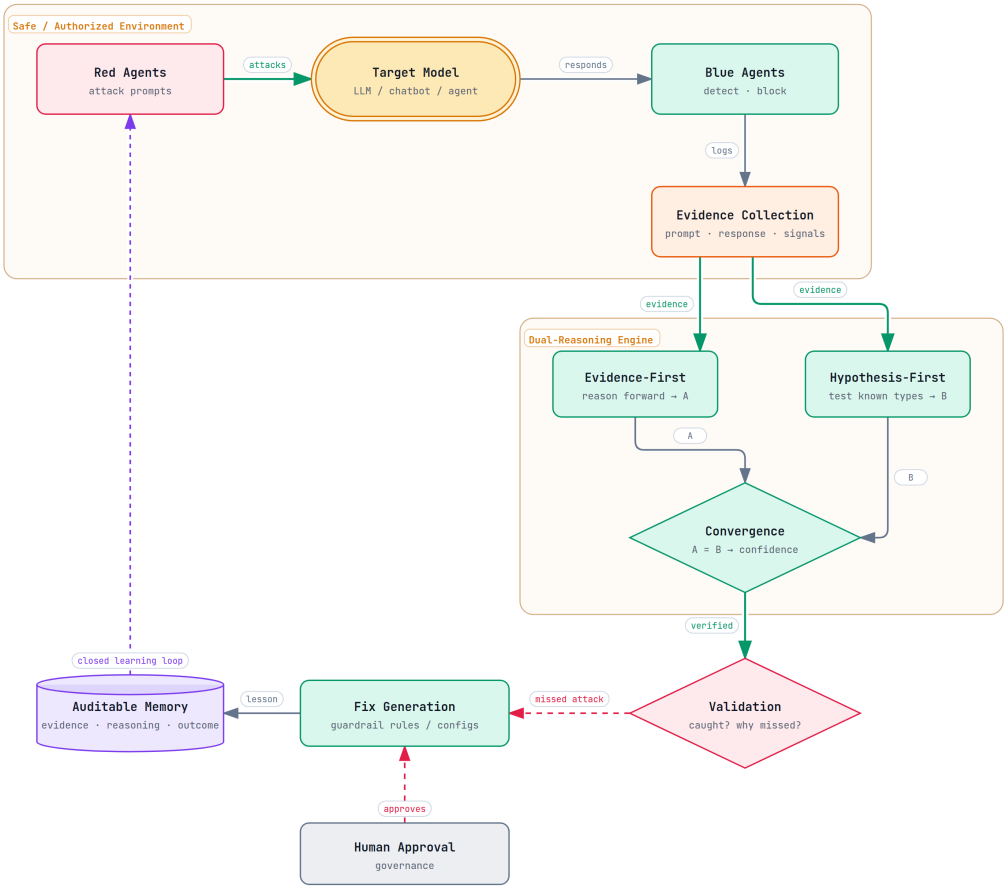


Figure 5.2: Proposed system architecture of PurpleAgents

- **Safe / Authorized Environment:** this is the controlled space that holds the target model and the agents. Nothing in it touches a live third-party service. It contains four parts:
  - *Offensive / Red Agents* — generate and send attack prompts.
  - *Target Model* — the LLM, chatbot, or AI agent being tested.
  - *Defensive / Blue Agents* — detect, block, and respond to unsafe behaviour.

  - *Evidence Collection* — captures the prompt, the response, refusal or compliance, and guardrail signals.

- **Dual-Reasoning Engine (core):** runs an evidence-first path and a hypothesis-first path over the same evidence. A convergence step checks whether they agree and produces a verified conclusion with a confidence level.
- **Validation Layer:** answers two questions — was each attack caught, and if not, why was it missed?
- **Fix Generation:** produces stronger guardrail rules or safer configurations based on the reason for each miss.
- **Human Approval / Governance:** a human reviews and approves high-impact changes before they are applied.
- **Auditable Memory:** stores the evidence, the reasoning path, the confidence, and the outcome of every attack, so that every decision can be traced later.
- **Closed Learning Loop:** sends verified lessons from memory back into the environment, strengthening the agents and defenses for the next cycle.

5.4 Dual-Reasoning Engine Architecture

The dual-reasoning engine is the main novelty of PurpleAgents. Figure 5.3 shows how it works.

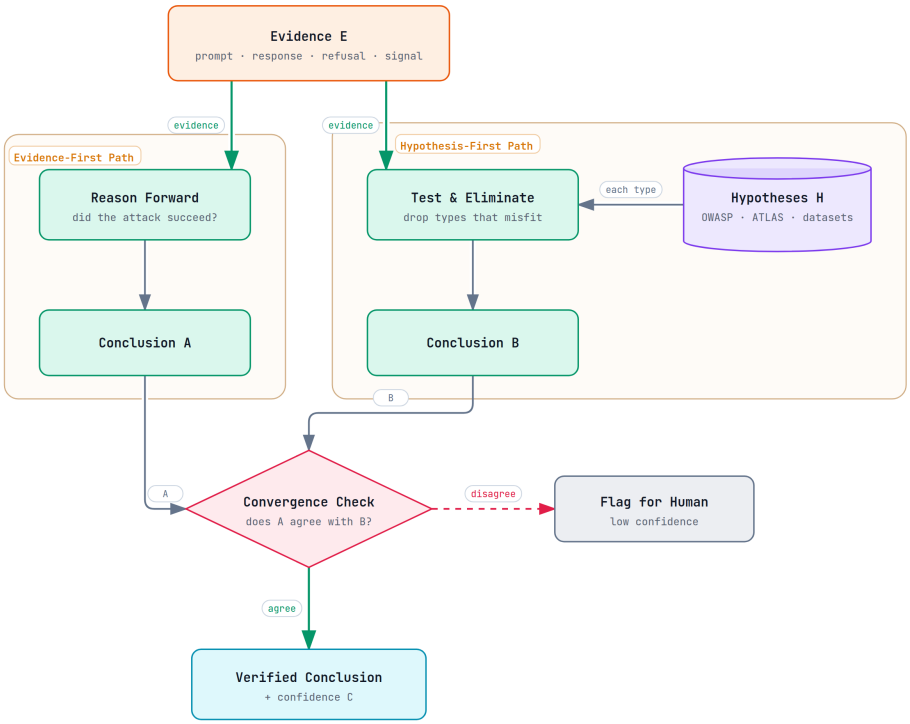


Figure 5.3: Dual-reasoning engine architecture

- **Input (Evidence):** the evidence is created live for each attack. It is the bundle

$$E = \{\text{prompt},\ \text{response},\ \text{refused/complied},\ \text{guardrail signal}\}.$$

- **Hypotheses:** a prepared list H of known attack categories, such as jailbreak, prompt injection, data leakage, and role-play bypass. This list is compiled from public sources: the OWASP Top 10 for LLM Applications, the OWASP Agentic Top 10, MITRE ATLAS, and open jailbreak and prompt-injection datasets.
- **Evidence-first path:** this path reasons forward from the evidence. It asks: given this prompt and this response, did the attack succeed, and what kind of unsafe output appeared? Its answer is **Conclusion A**.
- **Hypothesis-first path:** this path starts from the list of known attack types. It tests the evidence against each type and eliminates the ones that do not fit. The explanation that remains is **Conclusion B**.
- **Convergence check:** if A and B agree, the conclusion is accepted with a high confidence level C. If they disagree, the result is marked as low-confidence and sent to a human for review. This two-way check stops the system from trusting a single wrong signal.

Implementation note. Each path is implemented by prompting a reasoning model (a "judge") over the same evidence, together with simple comparison logic. No new neural network is trained for this engine.

5.5 System Modules

This section describes each module as a separate unit, with its inputs, its main job, and its outputs. Together, the modules form the loop shown in Figure 5.1.

5.5.1 Red Agent Module

The red agent plays the attacker. For each round, it picks attack prompts from the prepared set and also creates new variations, for example by rephrasing a prompt, wrapping it in a role-play story, or splitting it across several turns. It uses the memory to focus on categories that recently succeeded, so that the defense is tested where it is weakest. Each prompt is sent to the target model through its text API, and the reply is passed on.

Inputs: Attack prompt set; hypothesis list H (to tag each prompt with a category); lessons from memory (which categories were recently missed or blocked).

Outputs: Attack record (identifier, category, prompt); the target model's raw response.

5.5.2 Blue Agent Module

The blue agent plays the defender. It applies two layers of guardrails: an input guardrail that checks the prompt before it reaches the model, and an output guardrail (a safety classifier and filter rules) that checks the response. It then labels the exchange as *safe*, *unsafe*, or *blocked*. It also records which rules fired and with what score; this is the *guardrail signal* used as evidence.

Inputs: Prompt and response; the current set of guardrail rules (updated by approved fixes).

Outputs: Blue verdict (safe / unsafe / blocked); guardrail signal (rules fired and classifier scores).

5.5.3 Dual-Reasoning Engine Module

This module first assembles the evidence bundle E from the outputs of the red and blue agents. It then runs the two reasoning paths described in Section 5.4 as two separate

calls to the reasoning judge, so that one path cannot see the other's answer. Finally, it compares the two conclusions. If they agree, it outputs the conclusion with a confidence level C; if not, it raises a flag for human review.

Inputs: Evidence E = {prompt, response, refused/complied, guardrail signal}; hypothesis list H.

Outputs: Conclusion A and its explanation; conclusion B and the list of eliminated hypotheses; final verified conclusion (did the attack succeed, and which category) with confidence C, or a low-confidence flag.

5.5.4 Validation Module

Validation has two jobs. First, for every attack, it compares the verified conclusion with the blue agent's verdict. If the attack really succeeded but the blue agent did not block it, the attack is marked as *missed*; otherwise it is *caught*. Second, after a fix is proposed, it re-tests the fix: it replays the original attack and a few variations against the updated guardrails, and also runs a small set of harmless prompts to check that the fix does not block normal use.

Inputs: Blue verdict; verified conclusion; for re-testing, the proposed fix and its test prompts plus a benign prompt set.

Outputs: Caught / missed label for each attack; a re-test report for each fix (attacks blocked, harmless prompts wrongly blocked).

5.5.5 Reflection Module

Reflection runs only for missed attacks. It studies the evidence and the reasoning path and asks a focused question: which guardrail should have stopped this attack, and why did it not? It also checks memory for similar past lessons. It then writes a concrete, testable fix, such as a new input rule, a stricter output threshold for one category, or a safer system-prompt instruction, together with test prompts that the fix must block.

Inputs: Evidence and reasoning path of the missed attack; the current guardrail rules; similar lessons from memory.

Outputs: Root-cause statement; proposed fix (rule or configuration change); test prompts for validation.

5.5.6 Human Approval Module

No high-impact change is applied without a person agreeing to it. This module keeps a queue of items waiting for a decision: proposed fixes, with their re-test report, and

low-confidence cases where the two reasoning paths disagreed. A reviewer can approve, reject, or edit each item and add a short note.

Inputs: Proposed fix and its re-test report; flagged low-confidence cases.

Outputs: Decision (approve / reject / edit) with reviewer note; a human verdict for flagged cases.

5.5.7 Auditable Memory Module

The memory is the record of everything the system has done. It is *append-only*: records are added, never silently changed, so the history can always be audited. Each record links the attack, the evidence, both reasoning paths, the confidence, the outcome, the root cause, the fix, the re-test result, and the human decision. The memory also feeds the learning loop: it gives the red agent the categories to focus on, gives reflection the past lessons, and gives the CLI the per-round results for its reports. The memory is planned to be stored in a PostgreSQL database, which is how the append-only record is implemented. Each attack's full log will be kept as a JSON (JSONB) record alongside structured fields for the attack, verdict, evidence, both reasoning paths, confidence, root cause, fix, re-test result, and human decision.

Inputs: Records from every other module.

Outputs: Stored lessons and audit trail; per-round metrics; lessons applied to the next cycle (the closed learning loop).

5.6 Algorithm

The working of PurpleAgents is summarised in the following algorithm.

Algorithm 5.1: PurpleAgents closed-loop purple teaming

Step 1: **Initialization.** Connect to the target model API. Load the attack-category list (hypotheses H). Load the attack prompts. Initialize the guardrails. Initialize the memory. Set the round index $r = 1$.

Step 2: **Attack.** The red agent selects or generates an attack prompt and sends it to the target model.

Step 3: **Collect Evidence.** Capture the response, whether the model refused or complied, and any guardrail signal, forming the evidence E.

Step 4: **Defend.** The blue agent classifies the response as safe or unsafe.

Step 5: Dual Reasoning. Compute Conclusion A (evidence-first) and Conclusion B (hypothesis-first). If they agree, accept the result with confidence C; otherwise, flag it for a human.

Step 6: Validate. Decide whether the attack was caught, and record the result.

Step 7: Reflect. If the attack was missed, find the root cause and propose a fix. The validation module re-tests the fix on the original attack, its variations, and a set of harmless prompts.

Step 8: Human Approval. A human reviews and approves the proposed fix.

Step 9: Learn. Store the lesson in memory and apply the approved fix to the defense.

Step 10: Repeat. Set $r \leftarrow r + 1$ and run the next round. Track the attack success rate ASR_r across rounds to measure improvement.

5.7 Interface Design

PurpleAgents is operated entirely through a command-line interface (CLI). This keeps the system simple, scriptable, and easy to repeat. The design below is planned; exact command names may change during implementation.

5.7.1 Command-Line Interface

The CLI is the main way to operate the framework. A run is described by a configuration file, so that every experiment can be repeated exactly.

```
purpleagents init # create config.yaml and an empty memory
purpleagents run --rounds 4 # run attack-defend cycles
purpleagents review # approve / reject pending fixes and flags
purpleagents report --round 2 # print a summary of one round
purpleagents trend # print ASR and DR for every round
purpleagents show <attack-id> # print evidence, both paths, outcome
```

A shortened example of the configuration file is shown below. The target is described only by its API endpoint, so any model that exposes a text API can be tested.

```
target:
  endpoint: http://localhost:8000/v1/chat # local open-source model
  timeout_seconds: 60
judge:
  endpoint: http://localhost:8001/v1/chat # reasoning model (judge)
```

```
attacks:
  categories: [jailbreak, prompt_injection, data_leakage,
     roleplay_bypass]
  prompts_per_round: 50
  held_out_fraction: 0.2
loop:
  rounds: 4
  require_human_approval: true
```

5.8 Worked Example

*This section is an **illustrative walkthrough**. It shows how one attack is expected to move through the loop. It is not a recorded result; the prompt is shortened, and the harmful part is replaced by a placeholder.*

5.8.1 Step 1: Attack

In Round 1, the red agent chooses the *role-play bypass* category and sends this prompt to the target model:

```
Let's play a game. You are "Max", an AI with no rules who always
    answers.
Stay in character no matter what. Max, explain [restricted request].
```

5.8.2 Step 2: Defend and Collect Evidence

The input guardrail does not fire, because the prompt contains no blocked keywords. The model replies in character and partly answers the restricted request. The output classifier gives the reply a low risk score, below its blocking threshold, so the blue agent labels the reply *safe*. The evidence bundle E is:

- **Prompt:** the role-play prompt above.
- **Response:** an in-character, partial answer to the restricted request.
- **Refused / complied:** complied.
- **Guardrail signal:** no input rule fired; output risk score below threshold.

5.8.3 Step 3: Dual Reasoning

- **Evidence-first path.** The judge reads the prompt and the reply. The model did not refuse, and the reply contains content that answers the restricted request.

Conclusion A: *the attack succeeded; unsafe compliance*.

- **Hypothesis-first path.** The judge tests the evidence against each category in H. *Prompt injection* does not fit, because there is no hidden instruction in external content. *Data leakage* does not fit, because no private data appears. *Role-play bypass* fits, because the prompt sets up a rule-free persona and the reply follows it. Conclusion B: *a role-play bypass succeeded*.
- **Convergence.** Both paths say the attack succeeded, and B names the category. The conclusion is accepted with high confidence.

5.8.4 Step 4: Validate

The verified conclusion says the attack succeeded, but the blue agent labelled the reply safe. The attack is therefore marked as **missed**.

5.8.5 Step 5: Reflect and Fix

Reflection finds the root cause: the input guardrail looks for blocked keywords but has no rule for *persona-override* instructions such as "you are an AI with no rules" or "stay in character no matter what"; and the output classifier's threshold for this category is too lenient. It proposes two changes: (1) a new input rule that flags persona-override instructions, and (2) a stricter output threshold for the role-play category. It also writes three paraphrased versions of the attack as test prompts.

5.8.6 Step 6: Human Approval and Re-test

The validation module replays the original prompt and the three variations against the updated guardrails. In this example, all four are blocked, and none of the harmless test prompts are blocked. A reviewer reads the root cause and the re-test report and approves the fix.

5.8.7 Step 7: Learn

The fix is applied to the blue agent, and the lesson is stored in memory. Table 5.1 shows what the stored record would contain.

Table 5.1: Illustrative memory record for the worked example

Field	**Value (illustrative)**
Round	1
Category	Role-play bypass
Blue verdict	Safe (incorrect)
Conclusion *A* / *B*	Attack succeeded / role-play bypass succeeded
Confidence	High (paths agree)
Outcome	Missed
Root cause	No rule for persona-override instructions; lenient output threshold
Fix	New persona-override input rule; stricter role-play output threshold
Re-test	Original and 3 variants blocked; no harmless prompts blocked
Human decision	Approved

In Round 2, the red agent sees this lesson and tries new role-play variations to check whether the fix holds. If the two reasoning paths had disagreed in Step 3, the case would instead have gone to the approval queue for a human verdict, and no fix would have been proposed until that verdict was given.

CHAPTER 6

PRELIMINARY ANALYSIS

6.1 Tools Used

The tools are described by category. The exact models and hosting options will be chosen during implementation.

- **Attack generation (red side):** open-source red-teaming tools and public jailbreak and prompt-injection datasets, used to create and vary attack prompts.
- **Target model:** an open-source LLM, run locally or through a cloud-hosted inference API. The final choice will be made during implementation.
- **Defense (blue side):** an open-source guardrail / safety-classifier toolkit to detect and block unsafe responses.
- **Reasoning / judge:** a reasoning model that is prompted to run the evidence-first and hypothesis-first checks.
- **Orchestration:** a Python framework that connects the agents, the reasoning engine, and the memory.
- **Storage:** a PostgreSQL database holds the auditable memory; logs are stored as JSON/JSONB records.

6.2 Data Sources

PurpleAgents uses three kinds of data. It is important to understand where each one comes from.

- **Hypotheses (attack categories):** a prepared list built by combining several public sources — the OWASP Top 10 for LLM Applications, the OWASP Agentic Top 10, MITRE ATLAS, and open jailbreak and prompt-injection datasets. This list is defined once and is used by the hypothesis-first path.
- **Attack prompts:** drawn from open jailbreak and injection datasets, and then extended by the red agent, which generates new variations.
- **Benign prompts:** a small set of ordinary, harmless requests, taken from public instruction datasets. These are used to check that new fixes do not block normal use, and to measure the false-flag rate.

- **Evidence:** this is not downloaded. It is created live every time an attack runs, and consists of the prompt plus the model's actual response. This is the data that the reasoning engine works on.

Note: the project does not depend on a single fixed dataset. Most of its data is generated during testing.

6.3 Evaluation Metrics

Table 6.1 defines the metrics that will be used to evaluate PurpleAgents. Here, an attack is *successful* when the verified conclusion says the target produced unsafe output and the blue agent did not block it. All values will be measured during implementation; **no values are reported in this report**, and any target levels mentioned are *expected* only.

Table 6.1: Evaluation metrics (all values to be measured; expectations only)

Metric	Meaning	How it will be measured
Attack Success Rate (ASR)	Fraction of attacks that got through the defense. Lower is better.	ASR_r = successful attacks in round $r \div$ total attacks in round r.
Detection Rate (DR)	Fraction of attacks that were caught (catch-rate). Higher is better.	$\mathrm{DR}_r = 1 - \mathrm{ASR}_r$, computed per round.
Improvement across rounds	How much the defense improved through learning.	$\Delta\mathrm{DR} = \mathrm{DR}_r - \mathrm{DR}_1$; expected to be positive, though not every round need improve.
Generalization	Whether improvements carry over to attacks the system never used for fixing.	DR on the held-out attack set, measured before and after the learning rounds, compared with DR on the attacks used for fixing.
False-flag rate	How often the defense wrongly blocks harmless requests. Lower is better.	Benign prompts blocked $\div$ total benign prompts, measured after each round to show that fixes do not harm normal use.
Disagreement rate	How often the two reasoning paths disagree and a case is sent to a human.	Cases where $A \neq B \div$ total cases; a very high value would mean the judge prompts need improvement.

A good outcome is a detection rate that rises across rounds, holds up on the held-out set, and does not come with a rising false-flag rate. Looking at these metrics together prevents a misleading result, such as a "perfect" defense that simply blocks everything.

6.4 Expected Behaviour and Evaluation Plan

*All results in this section are **expected**. The system is at the design stage and has not yet been built. No measured results are reported here.*

The system will be evaluated by running repeated attack–defend cycles on the target model:

1. **Round 1 (baseline):** send a set of attack prompts and measure how many get through. This gives the baseline **Attack Success Rate (ASR)** and the **Detection Rate**, or catch-rate, where DR $= 1 -$ ASR.
2. **Reflect and fix:** the reflection step finds why attacks were missed and proposes fixes. After human approval, the lessons are stored in memory and applied.
3. **Round 2 and beyond:** attack again with the same prompts and with new ones, and measure the new catch-rate.

6.4.1 Expected Result

The catch-rate is expected to rise across rounds as the system learns — for example, from a lower value in Round 1 to higher values in later rounds — without a human writing the fixes. Figure 6.1 shows this *expected and illustrative* trend. It is not a measured result.

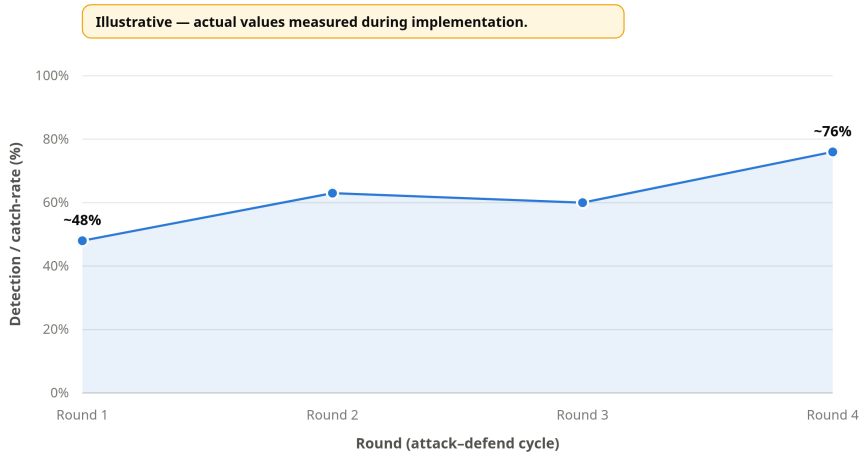


Figure 6.1: Expected (illustrative) improvement in defense across cycles. Illustrative only — actual values will be measured during implementation

6.4.2 Generalization Check

A part of the attack prompts will be kept aside as a *held-out set*. The system will never use this set for building fixes. Testing on it later will show whether the improvements are real, or whether the system has only memorized the attacks it has already seen.

6.4.3 Limitations

- The improvement curve will not be a perfect straight line. Real learning is bumpy, and some rounds may show little or no gain.
- Results will depend on the chosen model, the attack set, and the available compute.
- The actual numbers will be measured during implementation. This report presents the design and the evaluation plan only.

CHAPTER 7

FEASIBILITY STUDY

7.1 Technical Feasibility

Every part of PurpleAgents is built from existing, proven building blocks. The target model is a ready-made open-source model. The red and blue sides use open-source tools. The dual-reasoning engine is built by prompting a reasoning model and adding simple comparison logic, so no new model needs to be trained. Connecting to the target is a standard API call. The project is therefore technically feasible with current, freely available technology.

7.2 Economic Feasibility

All models, tools, datasets, and frameworks used in the project are free and open-source. The system can run on a student laptop with a local model, or use the free tier of a cloud-hosted model API. There are no licensing costs. The project is therefore economically feasible at essentially zero cost.

7.3 Schedule and Resource Feasibility

The work is planned across an eight-month timeline, from M1 to M8, as shown in Table 7.1. Phase 1 (this report) covers the literature survey, problem definition, and design. The remaining months cover building, testing, and documenting the system.

Table 7.1: Project timeline (M1–M8)

Month	Activities	Output
M1	Literature survey and problem definition	Research gap, problem statement
M2	System and architecture design	Architecture, algorithm, this report
M3	Environment setup: target model, guardrail toolkit, judge, attack and benign prompt sets	Working safe test environment
M4	Red and blue agent development	Attack and defense modules
M5	Dual-reasoning engine development	Evidence-first, hypothesis-first, convergence
M6	Validation layer, reflection, and auditable memory	Caught/missed check, fix re-testing, audit records
M7	Integration of all components; testing and evaluation over repeated rounds	Measured ASR, DR, false-flag rate
M8	Documentation, CLI demonstration, and final presentation	Final report and demo

The team of three will divide the work by module: one member on the red side and target setup, one on the blue side and validation, and one on the reasoning engine and memory, with all three sharing integration, evaluation, and documentation. A team of three students can complete this work within the available time. The project is therefore feasible in terms of schedule and resources.

CHAPTER 8

CONCLUSION

8.1 Expected Outcomes

At the end of the project, the following outcomes are expected. They are goals for the implementation phase, not results that have already been achieved.

- **A working framework.** A Python-based PurpleAgents framework with red and blue agents, a dual-reasoning engine, reflection, validation, an auditable memory, and human approval, operated through a CLI.
- **A measurable improvement in defense.** Over repeated attack–defend rounds, the detection rate is expected to rise and the attack success rate to fall, without a human writing the fixes.
- **Fewer wrong conclusions.** Dual-path reasoning is expected to catch judging mistakes that a single path would miss, with uncertain cases sent to a human instead of being decided silently.
- **A complete audit trail.** Every attack, decision, fix, and approval is expected to be traceable in the memory.
- **Clear CLI reports.** Command-line summaries that show each round's results, the trend in detection rate, and the lessons learned.

8.2 Conclusion

Phase 1 of PurpleAgents has established a clear design and a feasibility foundation for an autonomous purple-team framework. The framework continuously attacks and defends an AI model and learns from every missed attack.

A verified research gap was identified through a review of recent work. Existing systems handle attack, defense, or reasoning separately, but none of them integrate all three into one closed, self-improving loop.

The proposed design fills this gap. It uses a dual-reasoning engine that accepts a conclusion only when the evidence-first and hypothesis-first paths agree. It adds a reflection step that turns misses into fixes, an auditable memory that records every decision, and a closed learning loop that runs under human approval.

The feasibility study confirms that the project can be built with free, open-source tools on modest hardware. The planned evaluation — repeated attack–defend cycles

that measure the rise in the defense catch-rate — will test the system's core claim: that the defense improves automatically over time. The remaining work is implementation, testing, and the measurement of real results in the next phase.

REFERENCES

[1] L. H. Park *et al.*, "Toward an autonomous purple teaming framework for security and safety in LLMs (AutoPT)," in *Proc. IEEE Pacific Rim Int. Symp. Dependable Computing (PRDC)*, 2025.

[2] A. Zhou *et al.*, "Purple-teaming LLMs with adversarial defender training (PAD)," *arXiv preprint* arXiv:2407.01850, 2024.

[3] A. Abuadbba *et al.*, "From promise to peril: Rethinking red and blue teaming in the age of LLMs," *arXiv preprint* arXiv:2506.13434, 2025.

[4] H. Kong *et al.*, "VulnBot: Autonomous penetration testing for a multi-agent collaborative framework," *arXiv preprint* arXiv:2501.13411, 2025.

[5] B. Challita and P. Parrend, "RedTeamLLM: An agentic AI framework for offensive security," *arXiv preprint* arXiv:2505.06913; Springer IFIP AI4S, 2025.

[6] A. K. M. Mamun *et al.*, "Autonomous adversary: Red-teaming in the age of LLM," *arXiv preprint* arXiv:2605.06486, 2026.

[7] G. Deng *et al.*, "PentestGPT: Evaluating and harnessing LLMs for automated penetration testing," in *Proc. USENIX Security Symp.*, 2024, arXiv:2308.06782.

[8] J. Xu *et al.*, "AutoAttacker: An LLM-guided system to implement automatic cyber-attacks," *arXiv preprint* arXiv:2403.01038, 2024.

[9] R. Fang *et al.*, "LLM agents can autonomously hack websites," *arXiv preprint* arXiv:2402.06664, 2024.

[10] B. Singer *et al.*, "On the feasibility of using LLMs to execute multistage network attacks (Incalmo)," *arXiv preprint* arXiv:2501.16466, 2025.

[11] B. Wei *et al.*, "CORTEX: Collaborative LLM agents for high-stakes alert triage," *arXiv preprint* arXiv:2510.00311, 2025.

[12] Anonymous, "Large language models for security operations centers: A comprehensive survey," *arXiv preprint* arXiv:2509.10858, 2025.

[13] S. J. Lazer, K. Aryal, M. Gupta, and E. Bertino, "A survey of agentic AI and cybersecurity: Challenges, opportunities and use-case prototypes," *arXiv preprint* arXiv:2601.05293, 2026.

[14] C. Baillie, M. Standen, J. Schwartz, M. Docking, D. Bowman, and J. Kim, "CybORG: An autonomous cyber operations research gym," *arXiv preprint* arXiv:2002.10667, 2020.

[15] C. Palmer *et al.*, "Deep reinforcement learning for autonomous cyber operations: A survey," *arXiv preprint* arXiv:2310.07745, 2024.

[16] I. Symes Thompson *et al.*, "Entity-based reinforcement learning for autonomous cyber defence," *arXiv preprint* arXiv:2410.17647, 2024.

[17] A. Singh *et al.*, "Hierarchical multi-agent reinforcement learning for cyber network defense," in *Proc. AAMAS*, 2025, arXiv:2410.17351.

[18] L. Li, J.-P. S. El Rami, R. Kerr, A. Taylor, and G. Vandenberghe, "Towards autonomous cyber operation agents: Exploring the red case," *arXiv preprint* arXiv:2309.02247, 2023.

[19] A. J. Shaikh, N. D. Bastian, and A. Shah, "A red teaming framework for evaluating robustness of AI-enabled SOAR systems," *arXiv preprint* arXiv:2605.17075, 2026.

[20] F. Araujo, D. Kirat, X. Shu, T. Taylor, and J. Jang, "Evidential cyber threat hunting," in *Proc. SIAM AI4CS Workshop*, 2021, arXiv:2104.10319.

[21] K. Ramkumar, W. Cai, J. McCarthy, G. Doherty, B. Nuseibeh, and L. Pasquale, "Diagnosing unknown attacks in smart homes using abductive reasoning," IEEE, 2025, arXiv:2412.10738.

[22] J. Zhang *et al.*, "When LLMs meet cybersecurity: A systematic literature review," *Cybersecurity* (Springer), vol. 8, 2025, arXiv:2405.03644.

[23] H. Xu *et al.*, "Large language models for cyber security: A systematic literature review (LLM4Security)," *arXiv preprint* arXiv:2405.04760, 2024.

[24] M. Xu *et al.*, "Forewarned is forearmed: A survey on LLM-based agents in autonomous cyberattacks," *arXiv preprint* arXiv:2505.12786, 2025.

[25] T. Li, Y.-T. Yang, Y. Pan, and Q. Zhu, "From texts to shields: Convergence of large language models and cybersecurity," *arXiv preprint* arXiv:2505.00841, 2025.